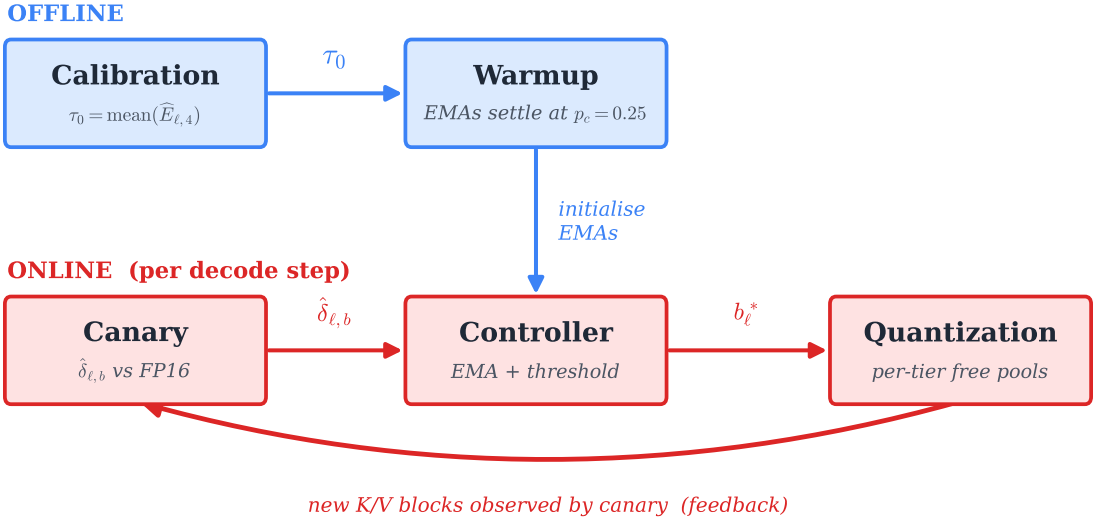
OFFLINE
Calibration
$\tau_0 = \mathrm{mean}(\widehat{E}_{\ell,4})$
τ_0
Warmup
EMAs settle at $p_c = 0.25$
initialise EMAs
ONLINE (per decode step)
Canary
$\hat{\delta}_{\ell,b}$ vs FP16
$\hat{\delta}_{\ell,b}$
Controller
EMA + threshold
b_ℓ^*
Quantization
per-tier free pools
new K/V blocks observed by canary (feedback)